Jacob Davis

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Education

Clemson University — **Clemson, SC**
Bachelor of Science in Computer Science and Mathematical Sciences (GPA: 3.97) — *Aug. 23 – May. 27*
- **Clubs & Affiliations**: Teaching Assistant (Data Structures and Algorithms), Honors Peer Mentor, Symphony Orchestra (Leader)
- **Relevant Coursework**: Operating Systems, Network Programming, Parallel Architecture, Quantum Computing, Databases
- **Technical Skills**: C++, Python, C, JavaScript, Go, SQL, Kubernetes, Docker, CUDA, JAX, PyTorch, OpenMP, OpenCL, Linux, Git

Experience

Google — **Sunnyvale, CA**
Software Engineer Intern — *May. 26 – Aug. 26*
- Conserved up to **50%** of SmartNIC buffer resources by engineering an allocation optimization in **C++** for a software-defined network.
- **Mitigated a production alert** caused by buffer exhaustion across high-density machines by developing a thread-safe and **low-latency** asynchronous callback to dynamically expand packet processing queues.
- Prevented resource depletion during feature downgrade through implementing safe state restoration and safeguard clamping upon **hitless transfer** within a cloud distributed system.

Clemson Research Computing and Data — **Clemson, SC**
Software Engineer Intern — *Jan. 26 – May. 26*
- Decreased website latency by **20%** by optimizing an OpenAI LLM service with local caching in **Go** to lower **MySQL** database reads.
- Enabled LLM performance evaluation under **hundreds** of configs in a **Kubernetes** and **Slurm** cluster by building a benchmark suite.

NASA Goddard Space Flight Center — **Greenbelt, MD**
Software Engineer Intern — *Jun. 25 – Jan. 26*
- Achieved a **5x performance increase** in a sky localization algorithm for compact binary mergers compared to **C** and **OpenMP** code on the CPU using **Python** and **JAX** to restructure the computation for the GPU.
- Reduced runtime by **99%** compared to sequential code by migrating open-source software used at gravitational-wave observatories to parallel processing frameworks using **CUDA**, **OpenCL**, and **OpenMP**.
- Presented at the 247th American Astronomical Society Meeting and won the "**Most Impactful Award**" at Clemson's HPC Day.

Clemson University International Center for Automotive Research — **Greenville, SC**
Software Engineer Intern — *Aug. 24 – May. 25*
- Achieved **95% accuracy** in differentiating trails and vegetation using a custom dataset of outdoor environments by training and fine-tuning LiDAR and image classification machine learning models in **Python** and **PyTorch**.
- Deployed data collection for an autonomous vehicle by developing a sensor data system, working with Applied Research Associates.

Cadence Design Systems — **San Jose, CA**
Application Engineer Intern — *May. 24 – Aug. 24*
- Pinpointed over **20 bugs** when testing the RTL code of a controller across **30+ features**, achieving **99%** functional coverage using **Universal Verification Methodology** (**UVM**), **SystemVerilog**, Cadence Xcelium, and SimVision.
- Increased workflow efficiency by **50%** by using **shell scripting** and **C++** to automate simulation analysis in a **Linux** environment.

Projects

High-Performance Cluster Computing — *Jan. 25 – Present*
- Reached a score of **~30 GFLOPS** on a custom four-node **Raspberry Pi** mini-cluster using **Spack**, **Docker**, and **OpenMPI** to simulate Clemson's **TOP500** Palmetto Cluster.
- Won **first place** in the interviews for a climate emulator at SC25's **international** Student Cluster Competition (SCC).
- Prepare a six-undergraduate team for SC26's SCC by directing weekly training sessions and system administration as **team lead**.

HPC Cluster System Admin Dashboard — *Jan. 26*
- Won the "**Best Use of MongoDB Award**" at HoyaHacks against **400+ participants** by engineering a real-time file monitoring system for HPC clusters featuring an interactive dashboard and terminal navigator using **JavaScript** and **Vue**.

Computer Vision Paper Synthesizer — *Sep. 25*
- Won the "**Best Musical Hack Award**" at HackMIT against **300+ teams**, achieving **90% key detection accuracy** to play a synthesizer on paper using **Python, YOLOv8**, and **MediaPipe**.

Asteroid Collision Simulator — *Sep. 25*
- Won the "**Most Inspirational Award**" at the NASA Space Apps Challenge's **third largest event** by building an asteroid collision simulator in **Godot** showcasing real-world seismic data and meteor impact visualizations.